

Name: _____

Date: _____



MARKING KEY

Year 10

Mathematics

Test 2, 2015

Topics – Algebra Skills

Total Time: 60 minutes

Weightings: 6% of the year.

Equipment

Allowed: 1/2 page notes (A4 one side)

64	%
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CALCULATOR FREE TEST

1. [11 marks: 1, 1, 1, 1, 2, 2, 2, 2]

Expand and where possible, simplify the following

a) $4(x+3)$

b) $2a(a+4)$

$= 4x + 12$

$= 2a^2 + 8a$

c) $-5b(3b-8)$

d) $(g+3)^2$ Short-cut $a^2 + 2ab + b^2$

$= -15b^2 + 40b$

$= (g+3)(g+3)$

$= g^2 + 3g + 3g + 9$ ①

$= g^2 + 6g + 9$ ①

e) $(x+5)(x-3)$

f) $(m-4)(2m+3) + 6m$

$= x^2 - 3x + 5x - 15$ ① $= 2m^2 + 3m - 8m - 12 + 6m$ ①

$= x^2 + 2x - 15$ ① $= 2m^2 + m - 12$ ①

g) $(x+5)(x-3)(2x+1)$

$= (x^2 - 3x + 5x - 15)(2x+1)$

$= (x^2 + 2x - 15)(2x+1)$ ①

$= 2x^3 + 4x^2 - 30x + x^2 + 2x - 15$

$= 2x^3 + 5x^2 - 28x - 15$ ①

-1-

2. [13 marks: 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2]

Factorise the following

a) $28y - 42$

$= 14(2y - 3)$

b) $6x + 18$

$= 6(x + 3)$

c) $x^2 + 4x$

$= x(x + 4)$

d) $3z^2 - 9z$

$= 3z(z - 3)$

e) $2(q+4) + p(q+4)$

$= (q+4)(2+p)$

f) $x^2 + 4x + 3$

$= (x+3)(x+1)$

g) $x^2 + x - 12$

$= (x+4)(x-3)$

h) $4x^2 + 36x + 80$

$= 4(x^2 + 9x + 20)$ ①

$= 4(x+4)(x+5)$ ①

i) $25 - 16y^2$

$= 5^2 - (4y)^2$ ①

$= (5+4y)(5-4y)$ ①

j) $(a+8)^2 - 49$

$= (a+8)^2 - 7^2$

$= (a+8+7)(a+8-7)$ ①

$= (a+15)(a-1)$ ①

-2-

3. [10 marks: 2.2.2.2.2.2]

Factorise the following (show your working out)

a) $3x^2 - 13x - 10$

b) $3a + 3b + ac + bc$

or other method

$$\begin{aligned} & 3(x-5) \quad \text{or other method} \\ & = \frac{(3x-15)(3x+2)}{3} \quad \text{①} \\ & = (x-5)(3x+2) \quad \text{①} \end{aligned}$$

c) $3a^2 - a^2b + 3ac - abc$

d) $5m + 5n + n^2 - m^2$

$$\begin{aligned} & = a^2(3-b) + ac(3-b) \quad \text{①} \\ & = (3-b)(a^2+ac) \quad \text{①} \\ & = 5(m+n) + (n+m)(n-m) \quad \text{①} \\ & = (m+n)(5+n-m) \quad \text{①} \end{aligned}$$

e) $9x^2 + 12x - 4 - y^2$ — QUESTION IS WRONG!

4. [9 marks: 1.1.1.2.2.2]

Simplify the following algebraic fractions

a) $\frac{2x}{3} + \frac{5x}{3}$

b) $\frac{2}{3} \times \frac{5x}{2} + \frac{x}{4}$

$$= \frac{7x}{3}$$

$$= \frac{10x}{4} + \frac{x}{4}$$

$$= \frac{11x}{4}$$

c) $\frac{2}{3} \times \frac{4}{3x} - \frac{1}{6x}$

d) $\frac{3}{2} \times \frac{x+2}{4} + \frac{2x-1}{3} \times \frac{4}{4}$

$$= \frac{8}{6x} - \frac{1}{6x}$$

$$= \frac{3(x+2)}{12} + \frac{4(2x-1)}{12} \quad \text{①}$$

$$= \frac{7}{6x}$$

$$= \frac{3x+6+8x-4}{12}$$

$$= \frac{11x+2}{12} \quad \text{①}$$

e) $\frac{(x-3)}{(x-3)} \times \frac{3}{x+2} + \frac{5x}{x-3} \times \frac{(x+2)}{(x+2)}$

f) $\frac{(x+4)}{(x+4)} \times \frac{x+2}{x+1} - \frac{2x-1}{x+4} \times \frac{(x+4)}{(x+4)}$

$$= \frac{3(x-3)}{(x-3)(x+2)} + \frac{5x(x+2)}{(x-3)(x+2)}$$

$$= \frac{(x+4)(x+2)}{(x+4)(x+1)} - \frac{(2x-1)(x+1)}{(x+4)(x+1)} \quad \text{①}$$

$$= \frac{3x-9+5x^2+10x}{(x-3)(x+2)} \quad \text{①}$$

$$= \frac{(x^2+2x+4x+8)-(2x^2+2x-1x-1)}{(x+4)(x+1)}$$

$$= \frac{5x^2+13x-9}{(x-3)(x+2)} \quad \text{①}$$

$$= \frac{-1x^2+5x+9}{(x+4)(x+1)} \quad \text{①}$$

5. [5 marks: 1, 1, 1, 1, 2]

Simplify the following algebraic fractions

a) $\frac{x}{5} \times \frac{15}{y}$

= $\frac{15x}{5y}$

= $\frac{3x}{y}$

c) $\frac{8}{5x} \div \frac{2}{20x}$

= $\frac{4}{15x} \times \frac{20x}{2}$

= 16

b) $\frac{18a}{4a} \times \frac{16c}{12b}$

= $\frac{18c}{3b}$

= $\frac{6c}{a}$

or any other way

d) $\frac{5(x+3)}{(x+1)(x-5)} \div \frac{(x+3)}{(x-1)}$

= $\frac{5(x+3)}{(x+1)(x-5)} \times \frac{(x-1)}{(x+3)}$

= $\frac{5(x-1)}{(x+1)(x-5)}$

6. [4 marks: 1, 1, 1, 1]

Solve each of the following equations (show your working out)

a) $\frac{x}{3} + 8 = 20$

= $\frac{x}{3} = 12$

= $x = 36$

c) $4x - 7 = 6x - 12$

-7 = 2x - 12

5 = 2x

$\frac{5}{2} = x$

$2\frac{1}{2} = x$

b) $4(2x + 3) = 36$

= $8x + 12 = 36$

$8x = 24$

$x = 3$

d) $3(5x - 4) = 2(6x + 3)$

$15x - 12 = 12x + 6$

$3x - 12 = 6$

$3x = 18$

$x = 6$

7. [12 marks: 2, 2, 2, 2, 2, 2]

Solve each of the following equations (show your working out)

a) $\frac{2x+3}{5} = 5$

$2x+3 = 15$

$2x = 12$

$x = 6$

b) $\frac{x+4}{5} - 6 = -2$

$\frac{x+4}{5} = 4$

$x+4 = 20$

$x = 16$

c) $\frac{5x}{4} + \frac{x+5}{4} = \frac{x-2}{5} \times \frac{4}{4}$

$\frac{5(x+5)}{20} = \frac{4(x-2)}{20}$

$5x+25 = 4x-8$

$x+25 = -8$

$x = -33$

d) $\frac{-2x-5}{3} - 6 = 1$

$\frac{-2x-5}{3} = 7$

$-2x-5 = 21$

$-2x = 26$

$x = -13$

e) $\frac{5}{3} \times \frac{-3(x-1)}{2} = \frac{2(x+6)}{5} \times \frac{2}{2}$

$\frac{-15(x-1)}{10} = \frac{4(x+6)}{10}$

$-15x+15 = 4x+24$

$15 = 19x+24$

$-9 = 19x$

$-\frac{9}{19} = x$

f) $\frac{(x+2)9}{(x+2)} \times \frac{5}{x-1} = \frac{3}{x+2} \times \frac{(x-1)}{(x-1)}$

$\frac{5(x+2)}{(x+2)(x-1)} = \frac{3(x-1)}{(x+2)(x-1)}$

$5(x+2) = 3(x-1)$

$5x+10 = 3x-3$

$2x+10 = -3$

$2x = -13$

$x = -\frac{13}{2}$

$x = -6\frac{1}{2}$

① - working out

① - answer

~END OF TEST~